

Week 3 — Training Day 2: Mission Operations Training

*ORBIT NORTH · Tue 18 August 2026 · uOttawa
Cyberrange*

GNG 3100 · SYS 5186

Summer 2026

Day at a glance

When	What	Output
09:00 – 10:00	Voice-loop discipline + GO/NO-GO protocol	Voice card learnt
10:00 – 11:30	Procedure walk-through: NOM-001 and CTG-003	Q&A captured
11:30 – 12:00	Shift handover practice (NOM-012)	Two handovers logged
13:00 – 14:00	Flight-dynamics workshop	Pass timing predicted
14:00 – 15:30	Dry-run 1 — nominal pass, no failures	Operator-shift logs
15:30 – 16:30	Dry-run 2 — nominal pass with one failure injected	Anomaly tickets, debrief
16:30 – 17:00	Operations training quiz	Quiz submitted

Setting: uOttawa Cyberrange. Two consoles set up by 09:00. Both dry-runs use the **Nominal Post-Commissioning** breakpoint (BP-1).

1. Voice-loop discipline

[Claude Code: 350 words. Cover the canonical phrasing used at NASA and ESA mission control. Include a one-page reference card with: addressing pattern (“FD, AOCS — go for star-tracker enable”), GO/NO-GO call-and-response, hot-mike rules, “loop discipline” expectations, the rule that the FD does not command directly. Cite ESA Operations Manual for Mission Control and the Mission Operations Voice Procedure standard from NASA APPEL — <https://appel.nasa.gov/>

Also provide a short EN/FR glossary of voice-loop callouts. uOttawa is bilingual; the team may use either language but must use one consistently per shift.]

1.1 STANDARD CALLOUTS

[Claude Code: produce a 12-row table — callout · meaning · response. Examples: “Comm check 1-2-3”, “Standby plus 30”, “TTC report”, “AOS in 2”, “Pass closed”.]

1.2 GO/NO-GO PROTOCOL

[Claude Code: 150 words. The FD polls each position; each position returns GO, NO-GO with reason, or HOLD with reason. The FD makes the decision. No commanding without an explicit FD authorisation.]

2. Procedure walk-through

2.1 NOM-001 PASS STARTUP

[Claude Code: walk students through procedure `procedures/nominal/startup.md` step by step. Show how each step maps to MCS UI actions. ~350 words plus annotated MCS screenshots.]

2.2 CTG-003 TTC LINK LOSS RECOVERY

[Claude Code: walk students through `procedures/contingency/ttc_link_loss.md`. Emphasise the detection/assessment/recovery loop. ~350 words plus annotated MCS screenshots showing the FDIR panel and link-status telemetry.]

3. Shift handover practice (NOM-012)

[Claude Code: 150 words. Use procedure `nominal/shift_handover.md`. Each team performs two handovers — outgoing FD briefs incoming FD using the NOM-012 template — and the instructor times them. Target: under 6 minutes.]

4. Flight-dynamics workshop

[Claude Code: 400 words plus three Python figures. Workshop exercises:

1. Read the orbit (TLE in `configs/eosat1/orbit.yaml`); compute period and rev/day; predict the next 24 h of Iqaluit AOS times.
2. Compute pass duration as a function of maximum elevation; show the typical 8–12 minute polar pass.
3. Explain eclipse: where it starts in the orbit, how long it lasts, what the power and thermal consequences are.

Figures (use the brand stylesheet from `outputs/sample_figures/`):

- Ground-track + Iqaluit and Troll visibility cones
- Pass-elevation profile vs time for a typical Iqaluit pass
- Eclipse fraction over a 24 h window]

5. Dry-runs

DRY-RUN 1 — NOMINAL PASS

- **Scenario:** BP-1 (Nominal Post-Commissioning).
- **Duration:** 90 min real time, 90 min sim time (1× speed).

- **Procedures expected:** NOM-001 → NOM-002 (imaging) → NOM-003 (data downlink) → NOM-009 (routine health check) → LOS.
- **Failure injection:** none.
- **Outputs:** operator shift log; observer team uses Communications peer-review worksheet only.

DRY-RUN 2 — NOMINAL PASS WITH ONE FAILURE

- **Scenario:** BP-1 with one easy failure injected ~T+30 min.
- **Recommended injection:** `eps_overcurrent.yaml` on the `payload` line (forces CTG-012 Overcurrent Response).
- **Operator pass criterion:** anomaly detected, isolated, and procedure CTG-012 executed; pass continues to LOS.
- **Outputs:** operator shift log + anomaly ticket; observer team uses Technical peer-review worksheet.

6. Operations training quiz

[Claude Code: produce a 15-question MCQ + short-answer paper, 30 minutes. Topics:

- Voice-loop discipline (3 Qs)
- MCS UI navigation (3 Qs)
- Telemetry interpretation — limits and SIDs (3 Qs)
- Procedure structure — what's in a procedure file (2 Qs)
- Anomaly response steps (2 Qs)
- Pass timing — AOS / TCA / LOS arithmetic (2 Qs)

Solutions on a separate page for the instructor pack.]

7. Expected reading for Wednesday

Expected reading before Wednesday LEOP day. Read sections 1 – 4 of the *LEOP Day Instructor Pack* student-facing extract (a redacted version of `sim_leop_instructor.md`). Approximate reading time 30 minutes. Read procedures **LEOP-001, LEOP-002, LEOP-006, LEOP-007** in full (`procedures/leop/*.md`). Approximate reading time 30 minutes.

8. Tools, files & references

SIMULATOR FILES REFERENCED

- `configs/eosat1/procedures/nominal/startup.md`
- `configs/eosat1/procedures/contingency/ttc_link_loss.md`
- `configs/eosat1/procedures/nominal/shift_handover.md`
- `configs/eosat1/procedures/leop/*.md` (for evening reading)
- `configs/eosat1/scenarios/eps_overcurrent.yaml`

OPEN REFERENCES

- NASA APPEL — Voice loop and mission ops standards — <https://appel.nasa.gov/>
- ESA Operations Mission Operations Concept — https://www.esa.int/Enabling_Support/Operations
- ECSS-E-ST-70-11C — Space segment operability — <https://ecss.nl/standards/active-standards/ecss-e-st-70-11c-space-segment-operability/>
- NASA Systems Engineering Handbook (SP-2016-6105 Rev 2) §8 — <https://www.nasa.gov/reference/systems-engineering-handbook/>
- CCSDS PUS standards (background) — <https://public.ccsds.org/Publications/All-Pubs.aspx>